Uncovering a Simple Relation between Stellar Mass and Observed Galaxy Properties

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Abstract

We extract the relationship between galaxy stellar mass and directly observed photometric properties—redshift, apparent magnitude, and colour—using machine learning and symbolic regression. Our analysis is based on a computer-generated mock galaxy catalogue that reproduces the redshift and magnitude distribution of a flux-limited survey, with ugriz photometry and photometric redshift estimates derived from BCNz2. Random Forest regression identifies the $z-$band absolute magnitude and the $i-z$ colour as the dominant predictors of stellar mass, together accounting for more than 95% of the predictive power at low redshift ($z < 0.5$). Building on this, symbolic regression recovers a simple analytical expression involving only these two observables, achieving high accuracy ($R^2 \approx 0.96$, $RMSE \approx 0.15dex$) with minimal loss compared to the full Random Forest model. The resulting formula provides a compact and interpretable estimator of stellar mass in flux-limited photometric surveys, particularly in the low-redshift regime.

1 Introduction

1.1 Background and Motivation

Estimating the stellar masses of galaxies is a fundamental task in astrophysics, as stellar mass provides a key tracer of galaxy formation and evolution across cosmic time. A variety of approaches have been developed to infer stellar masses, most prominently spectral energy distribution (SED) fitting techniques [3] [5] and empirical relations based on mass-to-light ratios or scaling relations with galaxy properties [2][17]. These methods have been widely applied in large spectroscopic and photometric surveys, such as GAMA and SDSS-IV MaNGA [4], and have established stellar mass as a cornerstone observable in galaxy population studies. Nevertheless, conventional approaches suffer from several well-known limitations: SED modeling relies heavily on assumptions about stellar population synthesis, star formation histories, and dust attenuation, which can lead to systematic biases; empirical relations are often calibrated in limited regimes and may not generalize robustly; and both methods are sensitive to photometric noise and become computationally expensive when applied to the vast datasets produced by modern surveys [12].

1.2 Challenges and Limitations

With the advent of large-scale photometric surveys such as the Sloan Digital Sky Survey (SDSS)[20] and the upcoming Vera C. Rubin Observatory Legacy Survey of Space and Time (LSST)[11], the volume and diversity of available photometric data have grown dramatically. While empirical relations between broad-band colors and stellar mass have long been proposed, observational noise and selection effects often obscure the intrinsic correlations, limiting their reliability in practice. To address these challenges, machine learning methods have been increasingly employed in astrophysical inference tasks, ranging from photometric redshift estimation [8] to transient classification [19], offering improved predictive performance compared to traditional techniques [1]. However, many of these approaches rely on "black-box" models such as deep neural networks, which, despite their accuracy, provide limited interpretability and thus fall short of yielding physical insight into the underlying stellar population processes [14].

1.3 Scientific Significance of Symbolic Regression

Symbolic regression (SR) offers a promising alternative by directly discovering analytic expressions that capture the underlying relations in the data. Unlike conventional regression techniques or black-box ma-

chine learning models, SR explicitly balances predictive accuracy and interpretability, yielding closed-form equations that are easier to validate scientifically. This approach resonates with the historical trajectory of scientific discovery, where interpretable laws such as Kepler's orbital laws and Planck's radiation law were distilled into compact mathematical forms, underscoring the enduring value of interpretable expressions in advancing physical understanding [16]. Modern developments have substantially enhanced the power of SR: early demonstrations revealed its ability to distill natural laws from experimental data [15], while physics-inspired frameworks such as AI Feynman [18] and neural-symbolic hybrids [7] have broadened its applicability. More recently, efficient implementations like PySR [6] have enabled fast and robust discovery of analytical models even in noisy, high-dimensional datasets, making SR a feasible tool for astrophysical applications.

1.4 Contributions of This Work

In this work, we apply symbolic regression to the estimation of galaxy stellar masses from broad-band photometric observables. Our method demonstrates that the intrinsic mass–magnitude relation, often obscured by observational uncertainties, can be effectively recovered in analytic form. The resulting equations reveal that the $z-$band absolute magnitude and the $i-z$ color dominate the predictive relation, consistent with longstanding astrophysical expectations that redder bands more reliably trace the older stellar population and thus the underlying stellar mass [21][10]. Beyond accurate mass estimation, the closed-form expressions derived by symbolic regression provide enhanced interpretability: they not only highlight the most informative photometric indicators but also offer new physical insights into the stellar mass–light connection. This combination of predictive power and transparency positions symbolic regression as a valuable tool for extracting interpretable astrophysical relations from large photometric surveys.

2 Methods

2.1 Data Sample

We use a mock galaxy catalogue designed to mimic a flux-limited photometric survey. The catalogue provides, for each galaxy, the cosmological (noise-free) redshift, apparent magnitudes in the five optical bands (u, g, r, i, z), and photometric redshift estimates derived using the BCNz2 code [9].

To ensure completeness, we impose a flux limit of $M_i < 22.5$,where the cut is applied in the i-band. This choice balances depth with photometric accuracy, producing a well-characterized sample suitable for statistical analysis. For comparison, we also examined a deeper cut of $m_i < 22.5$,but found that the fiducial $m_i < 22.5$,limit yields more stable results.

2.2 Data Sample

We define the absolute magnitude M_b in band b as

$$M = m - 5\log_{10} d_L - 25, \tag{1}$$

where m_b is the observed apparent magnitude and $D_L(z)$ is the luminosity distance at redshift z. This differs from the standard definition because we do not attempt to estimate the $k(z)$-correction.

Note on the Absolute Magnitude Definition: The absolute magnitude defined in Equation (1) is a *pseudo*-absolute magnitude, as it does not incorporate the k-correction that accounts for the redshifting of the spectral energy distribution. Consequently, this quantity is not a pure measure of the rest-frame luminosity in a given band. However, in our machine learning framework, we provide the model with complementary information—specifically, the photometric redshift estimates and the multi-band colors (e.g., $i-z$). These observables collectively encode the information about the galaxy's spectral shape and its redshift. Therefore, the model is capable of *implicitly learning* the necessary corrections. The pseudo-absolute magnitude M_z, in conjunction with color indices, thus serves as an effective proxy for the rest-frame luminosity and stellar population properties, enabling the model to recover the intrinsic stellar mass.

This dataset therefore provides the necessary observables—redshift, apparent *ugriz* magnitudes, and derived absolute magnitudes—for investigating the relation between stellar mass and photometric properties.

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2.3 Feature Construction

We focus on broad-band photometry in five optical bands (u, g, r, i, z). In addition to absolute magnitudes, we construct color indices such as $(i - z)$, which serve as key predictors of stellar mass. We also experimented with different forms of the redshift (e.g., z, z+1, $z^2, 1/(z + 1)$), but our analysis of the key features indicated that they are redundant for model fitting. Redshift information is incorporated through photometric estimates, and the analysis is primarily restricted to $z < 0.5$, where the relation is most stable and performance metrics plateau.

Throughout, we use $\log_{10} M$ to denote the base-10 logarithm of the stellar mass (i.e., no normalization by M_0 unless otherwise stated).

2.4 Machine Learning Models

We employ two complementary machine-learning approaches to recover the relation between stellar mass and observed galaxy properties. The first is Random Forest regression, a widely used ensemble method. A Random Forest consists of many decision trees, each trained on a random subset of the data and features. By averaging over the predictions of multiple trees, the method captures complex, non-linear relations while remaining robust against overfitting. Importantly, Random Forests provide a quantitative measure of feature importance, which allows us to identify the photometric bands and colors that carry the most predictive power for stellar mass.

The second approach is symbolic regression. Unlike black-box machine-learning methods, symbolic regression searches for explicit analytical expressions that fit the data. It uses evolutionary algorithms to iteratively combine mathematical operators into candidate formulas, evaluating each for both accuracy and simplicity. The resulting expressions strike a balance between predictive performance and interpretability, yielding compact equations that can be physically understood and easily applied to survey data.

2.4.1 Random Forest Regression

We implemented Random Forest (RF) regressors using the `scikit-learn` library. Given the large sample size, only minimal tuning was required to obtain stable results. We adopted 100 trees ($n_{\text{estimators}} = 100$) with parallel training enabled (`n_jobs=4`) and fixed the random seed for reproducibility (`random_state=42`). Other hyperparameters were left at their default values, as increasing the tree depth or ensemble size did not improve performance in our tests. Feature importance was quantified using the mean decrease in squared error across splits, providing a robust ranking of photometric bands and color indices.

2.4.2 Symbolic Regression

For symbolic regression (SR), we used the `PySR` package, which applies genetic programming to search for compact analytical models. The search employed a population size of 30 and ran for 50 iterations. Allowed binary operators included addition, subtraction, multiplication, and division. A parsimony coefficient of 0.01 was applied to penalize overly complex expressions, with the maximum expression size limited to 15 nodes. Computation was distributed over 8 processes with multithreading enabled. To ensure convergence, an early stopping criterion was set at 10^{-5} relative improvement. The models were ranked by R^2 and root-mean-square error (RMSE), and the final selected expression balances predictive accuracy with interpretability.

3 Results

3.1 Data Sample and Observational Limits

Our analysis begins with a characterization of the fundamental properties and selection effects in the observational dataset. As a first step, we verify the expected relation between cosmological redshift (z) and luminosity distance (d_L) in the mock sample. Figure 1 shows that the data follow the general increasing trend predicted by the Hubble law. A simple linear fit provides a reasonable approximation

but fails to capture the curvature of the relation, particularly at low redshift. This deviation is expected, since in a Λ CDM cosmology the luminosity distance is not exactly linear in redshift but depends on the cosmological parameters. Distances are expressed in units of Mpc/h.

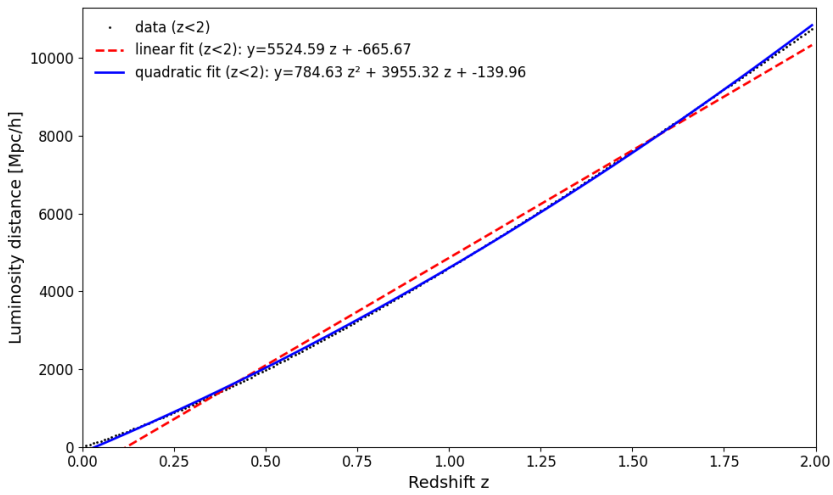


Figure 1: Hubble relation in the mock catalogue. Luminosity distance versus redshift for galaxies with 0¡z¡2. Distances are in Mpc/h. The red dashed line shows a linear fit, while the blue solid line shows a quadratic fit; the latter follows the expected curvature more closely (e.g. $RMSE_{linear} = 235.38 Mpc/h$ *vs.* $RMSE_{quadratic} = 42.12 Mpc/h$). This motivates the use of flexible models in subsequent analyses.

We restrict the fit to $0 < z < 2$, consistent with the catalogue limits, and express distances in Mpc/h. A linear model captures the overall trend but misses the expected curvature of $d_L(z)$ in a Λ CDM cosmology. We therefore also fit a quadratic polynomial, which substantially reduces the residuals (see RMSE in the caption), providing a closer empirical approximation within this redshift range.

For illustration, we compare two flux-limited subsamples defined by different i-band cuts: a deeper selection with $m_i < 23$, which shows a sharp truncation in the magnitude distribution, and a shallower selection with $m_i < 22.5$. In the latter case, the distribution peaks around $m_i \sim 22$ and then declines toward fainter magnitudes, so the imposed flux limit does not appear as a sharp edge in the histogram.

This turnover is the characteristic signature of a survey flux limit: beyond this point, the detection efficiency decreases rapidly because of instrumental sensitivity. Following the standard practice in studies using this dataset, we therefore adopt $m_i < 22.5$ as the completeness limit. This selection yields a well-characterized and statistically robust sample for quantitative analysis, predominantly excluding galaxies whose detection is severely affected by selection effects. Figure 2 illustrates this rationale.

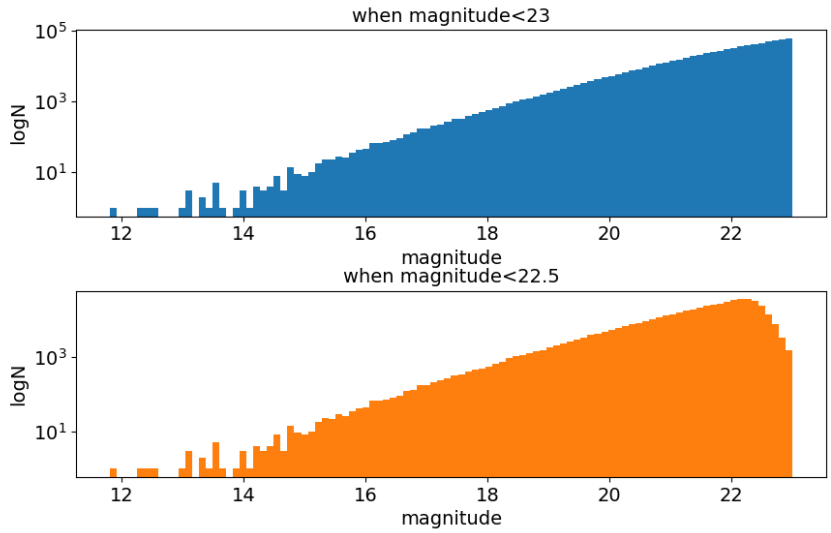


Figure 2: i-band apparent magnitude distributions for two flux-limited subsamples. Top: galaxies selected with $M_i < 23$, showing a sharp cut at the imposed flux limit. Bottom: galaxies selected with a stricter cut of $M_i < 22.5$. In this case the distribution peaks around $M_i \sim 22$ and declines toward fainter magnitudes, so the truncation at 22.5 is less visually sharp.

Figure 2 shows the relation between stellar mass and apparent i-band magnitude for galaxies restricted to $m_i < 22.5$. As expected, more massive galaxies tend to appear brighter, while the flux limit constrains the faint end of the distribution and ensures that the sample remains statistically complete within this range. The imprint of this flux limit is also evident in the redshift distribution of the same sample, shown in Figure 3. The number of galaxies rises to a pronounced peak at $z \sim 0.5$, with approximately 16,000 objects per bin, before declining at higher redshifts. This behavior reflects the survey's selection function: at low z, the small comoving volume contains few galaxies; the counts increase as the survey.

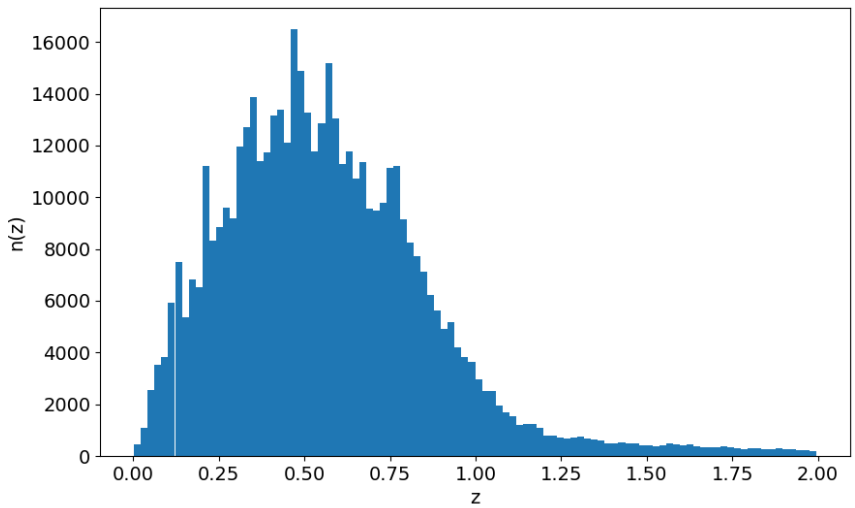


Figure 3: Stellar mass as a function of apparent i-band magnitude for galaxies with $m_i < 22.5$. The clear trend reflects the underlying correlation between brightness and stellar mass in the mock catalogue. The imposed flux limit ensures that the sample remains statistically complete while excluding the faintest galaxies that would otherwise suffer from severe selection effects.

3.2 Revealing the Underlying Mass-Magnitude Relation

To quantify the impact of observational uncertainties on our ability to recover intrinsic physical parameters, we compare three distinct data representations:

1. BCNz: Photometric redshift point estimates derived using the SED template-fitting code BCNz [13], which incorporates the inherent uncertainties from the observed broadband fluxes.
2. Perturbed: The final data product designed to mimic real observational catalogs, where additional perturbations have been applied to the BCNz estimates to simulate the cumulative effect of end-to-end measurement errors.
3. Cosmological: The ground-truth redshifts and derived physical parameters, representing the intrinsic, noise-free properties of the galaxies.

In order to realistically assess the robustness of the symbolic regression models, we constructed a perturbed version of the catalogue that incorporates typical observational errors. Two main sources of uncertainty were considered:

1. **Photometric errors:** Random noise was added to the photometry to mimic observational uncertainties. This includes photon-counting noise (Poisson statistics) as well as instrumental contributions such as readout noise from CCD detectors. These perturbations reproduce the scatter typically present in large photometric surveys.
2. **Photometric redshift errors:** In addition to flux uncertainties, the photometric redshifts were perturbed according to their estimated error distributions. This is crucial because redshift errors propagate into derived rest-frame magnitudes and colors, which directly affect the mass estimation. The distribution of perturbed redshifts is illustrated in Fig. 4, highlighting the broadening of the $z_{\rm phot}$–$z_{\rm spec}$ relation after introducing noise.

By comparing the performance of symbolic regression models on the original and perturbed catalogues, we are able to evaluate both the predictive accuracy and the stability of the recovered relations under realistic survey conditions. This provides a more reliable estimate of how the derived analytic expressions would perform when applied to observational data.

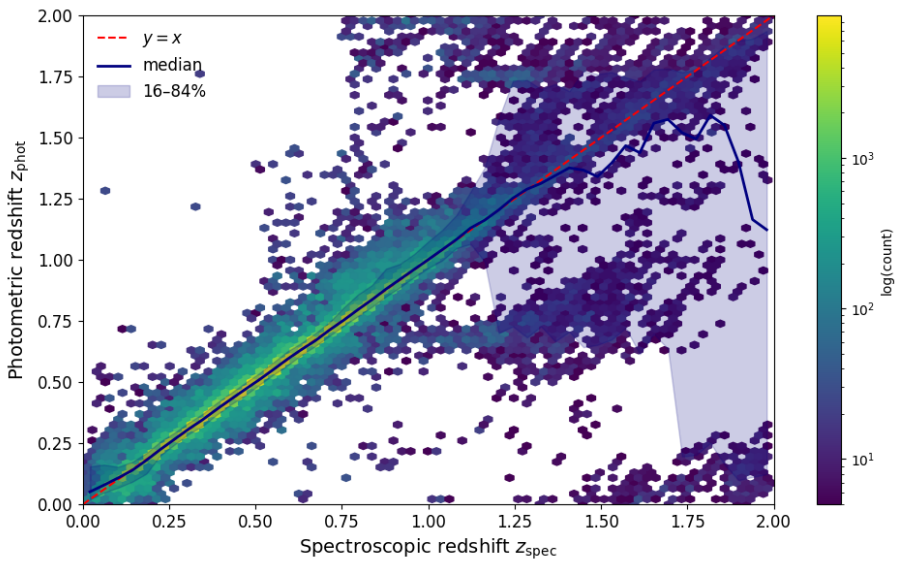


Figure 4: Comparison between spectroscopic and photometric redshifts in the perturbed catalogue.Colours show point density (hexbin, log count). The dashed line indicates $y = x$.To mitigate overplotting and quantify the scatter, we overlay the running median (solid line) and the 16–84 percentile range (shaded band) as a function of $z_{\rm spec}$. The distribution lies tightly along the one-to-one relation at low redshift and broadens toward higher redshift, as expected for photometric redshifts.

Figure4 compares the spectroscopic redshift $z_{\rm spec}$ with the photometric estimates $z_{\rm phot}$ in the perturbed catalogue. The density of galaxies is shown using a logarithmic hexbin scale, which avoids the

severe overplotting of individual points. The majority of objects follow the one-to-one relation ($y = x$, red dashed line), demonstrating that the simulated photometric redshifts recover the true redshifts with reasonable accuracy. To quantify the scatter, we overlay the running median (blue solid line) together with the 16–84 percentile range (shaded region). The distribution remains tightly clustered around $y = x$ at low redshift and gradually broadens toward higher z, consistent with the expected behaviour of photometric redshift uncertainties.

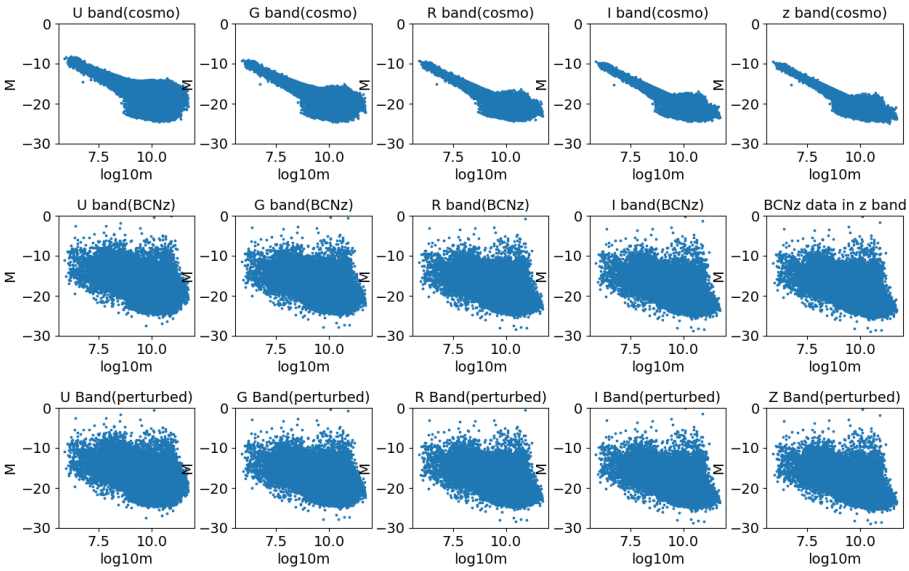


Figure 5: Stellar mass–magnitude relations across multiple photometric bands for the three catalogues considered in this work: **top row:** intrinsic relations from the Cosmological catalogue (noise-free, ground-truth values); **middle row:** relations from the BCNz catalogue, where photometric redshifts are estimated using SED template fitting; **bottom row:** relations from the Perturbed catalogue, which includes additional noise in both photometry and photometric redshift. The comparison illustrates how observational uncertainties progressively obscure the underlying tight correlations present in the cosmological data.

The pivotal insight emerges in Figure 5, where we examine the relationship between galaxy stellar mass ($\log_{10} M$) and absolute magnitude across all five bands (u, g, r, i, z). This comparison starkly contrasts the data in their different states:

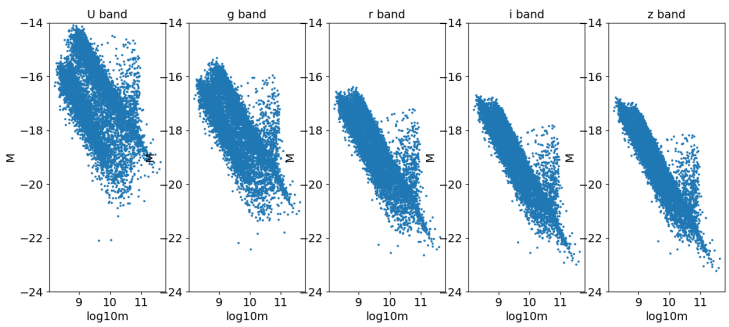
The Perturbed data—our proxy for a real observational catalog—forms a diffuse, amorphous cloud. In this representation, any underlying physical correlation is completely obscured by the compounded observational noise. The BCNz data, while also scattered, exhibits a marginally more concentrated distribution, as it constitutes a best-fit estimate prior to the final perturbation step. Nevertheless, the fundamental physical relation remains largely concealed.

In dramatic contrast, the Cosmological data reveal a sharp, coherent, and non-linear structure. This distinct "ladle-shaped" relation is the intrinsic astrophysical law governing galaxy mass and luminosity that our analysis seeks to recover.

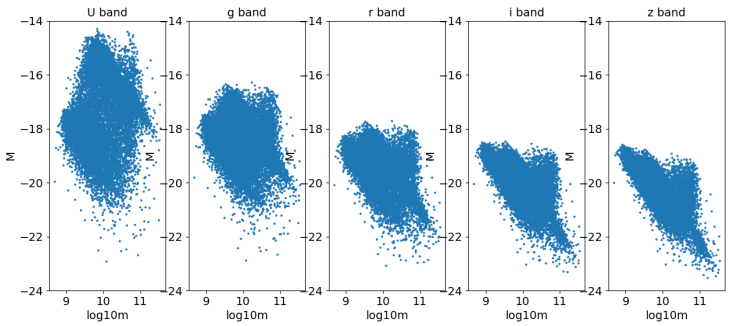
The stark contrast between the obscured relations in the Perturbed/BCNz data and the clear signal in the Cosmological data defines the central challenge of this work: a strong physical signal present in the universe is heavily contaminated in our observations. The primary goal of our machine learning approach is, therefore, to demystify this cloud of points and recover the hidden Cosmological relation directly from the noisy Perturbed data alone.

3.3 Redshift and Bandpass Evolution of the Relation

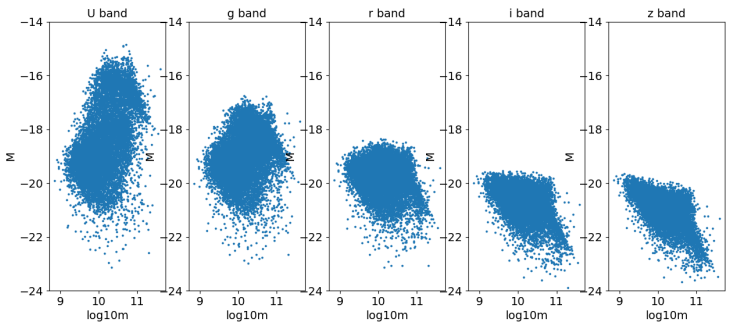
The distinct "ladle" structure uncovered in the cosmological data (5) is not a static relation. To investigate its dependence on cosmic time and the physical processes probed by different wavelengths, we dissect the data by both redshift and photometric bandpass.



(a) Panel (a), $z \sim 0.25$: The bifurcated, "ladle" structure is most pronounced, appearing as two distinct sequences across all bands, though markedly sharper in the redder (i, z) bands.



(b) Panel (b), $z \sim 0.5$: An intermediate state where the dual sequences begin to blur and merge, showing increased scatter, especially in the bluer (u, g) bands.



(c) Panel (c),$z \sim 0.75$: The coherent structure dissolves into a largely amorphous cloud, with the relation being most obscured in blue bands and only a faint hint of structure remaining in the z-band. The systematic evolution and bandpass dependence guide our feature selection for machine learning analysis.

Figures 6a,6b, and 6c present the mass-magnitude relation across all five ugriz bands for galaxies within narrow redshift slices centered at $z \sim 0.25$, 0.5, and 0.75, respectively. This multi-panel view reveals two profound and interconnected evolutionary trends.

3.3.1 Redshift Evolution: A Signature of Cosmic Maturation

A direct comparison across the three redshift slices reveals a dramatic evolution in the clarity of the ladle-shaped structure. As detailed in Figure6a, at low redshift ($z \sim 0.25$), the bifurcation into two distinct sequences is most pronounced across all bands. By intermediate redshift ($z \sim 0.5$, 6b), these sequences begin to blur and merge, showing increased scatter. Finally, at high redshift ($z \sim 0.75$, 6c), the coherent structure largely dissolves into an amorphous cloud. This evolution is a compelling signature of galaxy maturation over cosmic time. At earlier epochs (high z), galaxies are in a more chaotic state of assembly and rapid star formation, which prevents the establishment of a tight mass-magnitude sequence. As the universe ages (low z), galaxies settle into more stable configurations, allowing the clear physical relation to emerge and sharpen.

3.3.2 Bandpass Dependence: Tracing Star Formation versus Stellar Mass

Furthermore, within each redshift slice, the tightness of the relation is strongly dependent on the observed photometric band. Examining any single column in Figures 4a-c, we observe a systematic trend: the two sequences converge and the scatter substantially diminishes as we move from the bluer (u, g) to the redder (r, i, z) bandpasses. The relation is consistently most compact and well-defined in the z-band.

This bandpass dependence provides a crucial physical clue. Bluer bands are dominated by light from young, hot, massive stars, which are short-lived and trace recent, potentially stochastic, star formation episodes. Redder bands, however, are sensitive to the older, stable stellar populations that constitute the bulk of a galaxy's stellar mass. The fact that the mass-magnitude relation is sharpest in the reddest bands indicates that it is fundamentally a relation between a galaxy's stellar mass and the light from its underlying, quiescent stellar population. The increased scatter in blue bands is introduced by the transient and variable contribution of recent star formation.

These two findings—the strong redshift evolution and the systematic bandpass dependence—collectively guide our subsequent machine-learning analysis. They indicate that the cleanest physical signal resides in low-redshift galaxies (e.g., $z \sim 0.25$) and is best traced by the z-band magnitude and colors involving red bands (e.g., i-z). This insight directly informs our feature selection and the optimal redshift range for the final model derivation in the next section.

3.4 Machine Learning Recovery of the Physical Formula

Methodological Preliminaries: All machine learning analyses in this section were conducted using a consistent 4:1 train-test split ratio. The feature set comprised 13 parameters: absolute magnitudes in the ugriz bands (M_u, M_g, M_r, M_i, M_z), their derived colors (color_ug, color_gr, color_ri, color_iz), and four redshift-related terms (z, z^2, z-1, 1/[z+1]) to capture potential linear cosmological dependencies. These color indices serve as sensitive probes of stellar populations: colorug (u-g) traces recent star formation and UV continuum; color_gr (g-r) and color_ri (r-i) are sensitive to the Balmer break and overall stellar population age; color_iz (i-z) is particularly valuable for redshift estimation and dust extinction; and color_ig (i-g) provides additional leverage on stellar age and metallicity.The target variable was the logarithm of stellar mass, $\log_{10} M$. Random Forest models were configured with 100 estimators and utilized all features, while the subsequent symbolic regression operated on a refined subset to enhance efficiency.

3.4.1 Systematic Diagnosis via Random Forests - Feature Dominance and Evolution

Feature	Importance
M_z	0.794
color_iz	0.163
color_ug	0.013
color_gr	0.008
color_ri	0.008
redshift4	0.003
M_g	0.003
redshift2	0.002
redshift	0.002
redshift3	0.002
M_u	0.001
M_r	0.001
M_i	0.001

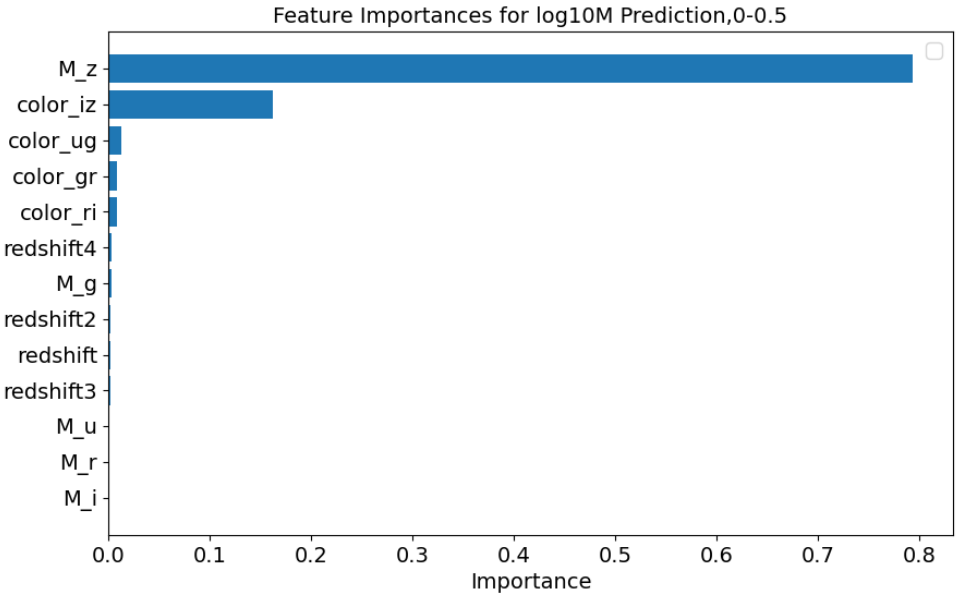
Table 1: Random Forest Feature Importance ($0 < z < 0.5$).

Feature	Importance
color_iz	0.455
M_z	0.231
color_gr	0.123
color_ug	0.072
color_ri	0.028
M_u	0.023
M_r	0.017
M_g	0.015
M_i	0.010
redshift4	0.009
redshift2	0.006
redshift3	0.006
redshift	0.006

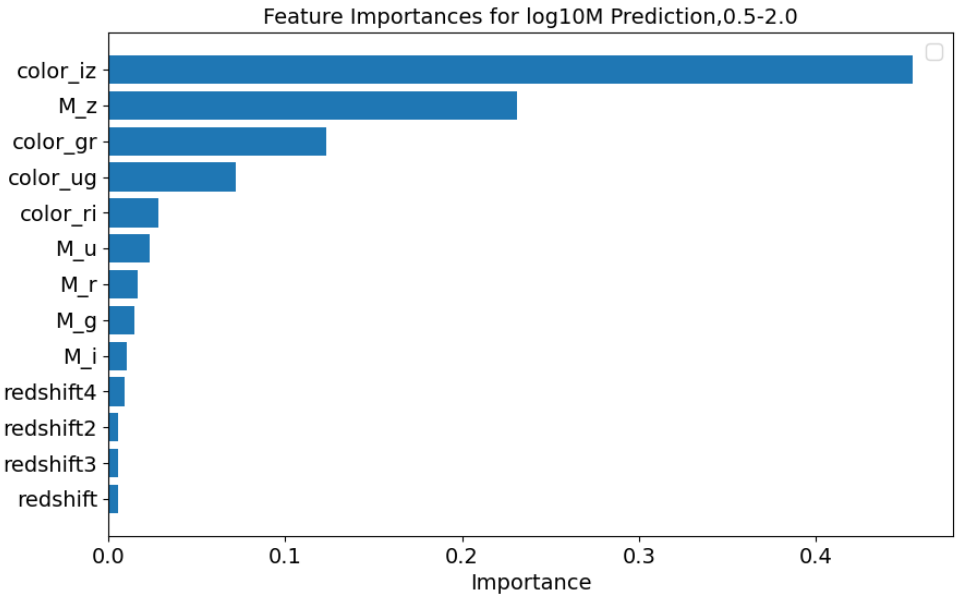
Table 2: Random Forest Feature Importance ($0.5 < z < 2.0$).

Table 3: Side-by-side comparison of RF feature importance at two redshift ranges. Values are rounded to three decimal places; small differences below ∼0.01 are not physically significant, and the interpretation should focus on the relative ranking and dominant features.

Our first objective was to identify the most predictive observables for stellar mass across cosmic time. We began this diagnostic in the low-redshift universe ($0 < z < 0.5$), where 3indicated the clearest physical signal. A Random Forest regressor trained in this regime revealed an extreme concentration of predictive power. As quantified in 1, two features overwhelmingly dominate: the z-band absolute magnitude (M_z) and the i-z color (color_iz). Together, they account for over 95% of the total feature importance (0.794 +



(a) Bar plot of feature importances for the $0 < z < 0.5$ bin, visually reinforcing Table 1.



(b) Bar plot of feature importances for the $0.5 < z < 2.0$ bin, visually reinforcing Table **??**.

Figure 7: Visualization of the Random Forest feature importance rankings in two redshift intervals. (A) At low redshift ($z < 0.5$), predictive power is overwhelmingly concentrated in M_z and color_iz. (B) At high redshift ($z > 0.5$), the importance is distributed across more features, and color_iz surpasses M_z as the top predictor.

0.163 = 0.957). The remaining 11 features contribute negligibly, with none exceeding 1.3% importance. This stark hierarchy, visually summarized in 7a, indicates that in this stable physical regime, a galaxy's stellar mass is almost entirely encoded in its z-band luminosity and its i-z color.

A starkly different picture emerges at high redshift ($0.5 < z < 2.0$). As 2 and 7b illustrate, while color_iz and M_z remain the top two features, their combined dominance drops to approximately 69% (0.455 + 0.231). Their individual rankings also swap, with color_iz becoming the primary predictor. Concurrently, the importance of other colors (e.g., color_gr, color_ug) and bluer absolute magnitudes rises substantially. This dilution and reordering reflect the increasingly complex and chaotic nature of galaxy assembly in the early universe, where a simple, tight relation is obscured by varied star formation histories and dust effects.

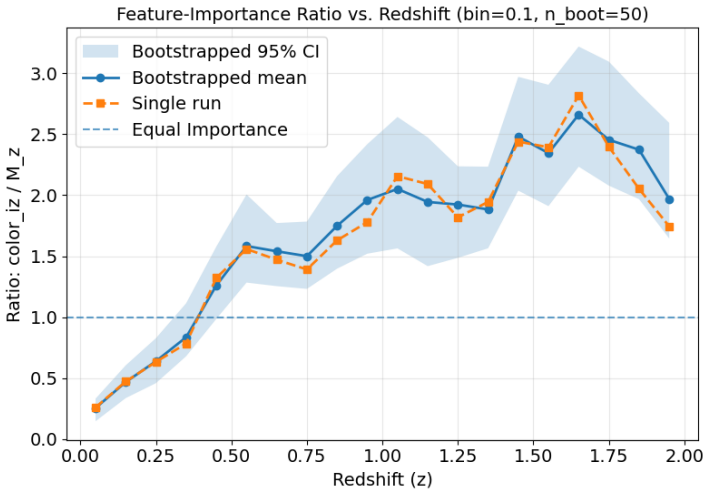


Figure 8: Combined panel: bootstrapped mean (solid line) with 95% CI (shaded), overlaid with the single-run result (dashed line with squares). This allows one to directly assess whether the answer from panel (a) falls within the scatter of panel (b).

To facilitate a direct comparison, we combine the single-run estimate with the bootstrapped statistics in a single panel (Fig. 8). For each redshift bin, we report the bootstrapped mean ratio mean[FI(color$_{iz}$)/FI(M_z)] together with its 95% confidence interval, and overlay the single-run result obtained with a fixed random seed. This layout allows us to verify at a glance whether the single-run estimate is consistent with the variability expected from data resampling.

As shown in Fig. 8, the single-run curve (dashed line with square markers) generally lies within the shaded 95% confidence band of the bootstrapped distribution across most redshift bins. Where the single-run falls well inside the band, the conclusion that color$_{iz}$ is comparatively less/more informative than M_z is robust to sampling variation. Occasional excursions outside the band typically occur in bins with fewer galaxies and wider uncertainty; we therefore avoid over-interpreting small deviations ($\lesssim 0.01$ in the ratio) at those redshifts.

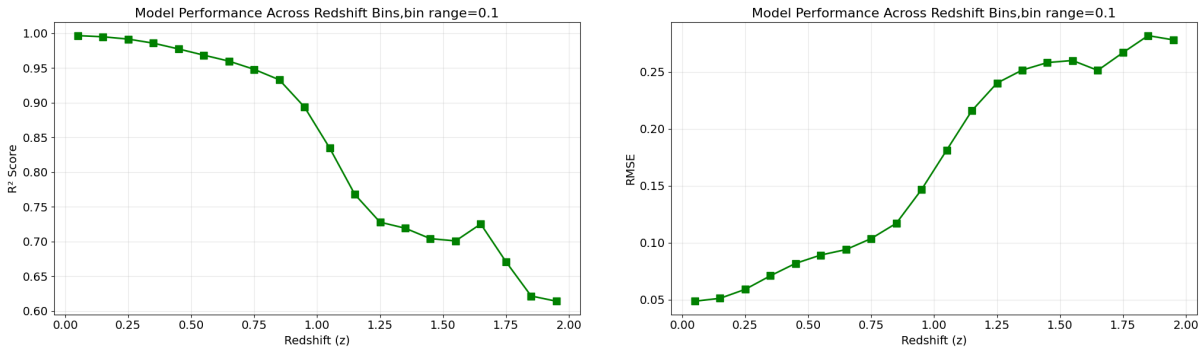
We interpret feature-importance ratios in conjunction with predictive metrics (R^2 and RMSE; definitions given where Fig. 9 is discussed), emphasizing trends that are simultaneously supported by stable rankings and by consistent out-of-sample performance. In practice, we focus on persistent patterns across bins rather than minute decimal differences, which are not physically meaningful at the scale of the bootstrap variability.

3.4.2 Dependence of performance on redshift.

Having established the dominance of M_z and color_iz as the principal features, we next sought to objectively determine the redshift domain within which a robust and reliable mass estimation relation could be derived. The evolution of the Random Forest model's performance, as depicted in 9, provides a clear, data-driven answer.

We evaluate performance using the coefficient of determination R^2 and the root-mean-square error (RMSE):

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2}, \qquad \text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}.$$



(a) Evolution of R^2 (bin width $\delta z = 0.2$, $n_{\text{boot}} = 50$).

(b) Evolution of RMSE (same setup as 9a).

Figure 9: Performance of the Random Forest stellar mass estimator across redshift bins.

3.4.3 Dependence of performance on redshift

Figure 9 shows the predictive performance of the Random Forest stellar mass estimator as a function of redshift, using a fine binning of $\Delta z = 0.1$. Two complementary metrics are shown: the coefficient of determination R^2 and the root-mean-square error (RMSE).

At low redshift ($z \lesssim 0.5$), the model achieves nearly perfect predictive accuracy, with $R^2 \gtrsim 0.98$ and RMSE below 0.1 dex. As redshift increases, both metrics degrade steadily: RMSE rises to ~ 0.25 dex by $z \approx 1.5$–2.0, while R^2 falls to below 0.7. This reflects the combined effect of fainter magnitudes, larger photometric uncertainties, and the intrinsic broadening of the mass–luminosity relation at higher redshift.

The finer binning reveals small fluctuations in the trend (e.g., around $z \sim 1.6$–1.7), which we interpret as statistical noise due to the reduced sample size per bin rather than genuine physical features. Nevertheless, the overall monotonic degradation of performance with redshift is robust.

Our selection of the redshift range $0 < z < 0.5$for the final derivation of the analytical formula is firmly grounded in this performance diagnosis, guided by three key principles:

1. Performance Stability: The interval $z < 0.5$ constitutes a distinct "plateau" of stable, high-fidelity performance. Operating within this regime ensures that our model's predictions are minimally affected by redshift-dependent systematic errors, maximizing the reliability of the resulting physical formula.

2. Astrophysical Context: The degradation of performance at higher redshifts is astrophysically expected. The observational data increasingly probe epochs of intense galaxy assembly, where complex star formation histories, strong dust attenuation, and greater observational uncertainties act to obscure the clean mass-magnitude relation evident at later cosmic times.

While it is noteworthy that other specific sub-intervals within $z < 0.5$ may exhibit marginally better performance, the choice of the full range from 0 to 0.5 is strategic. It provides a substantial sample size for robust model training while fully encompassing the stable performance plateau, offering a comprehensive representation of the local universe where the fundamental physical relation is most accessible.

3.5 Derivation of the Final Physical Formula

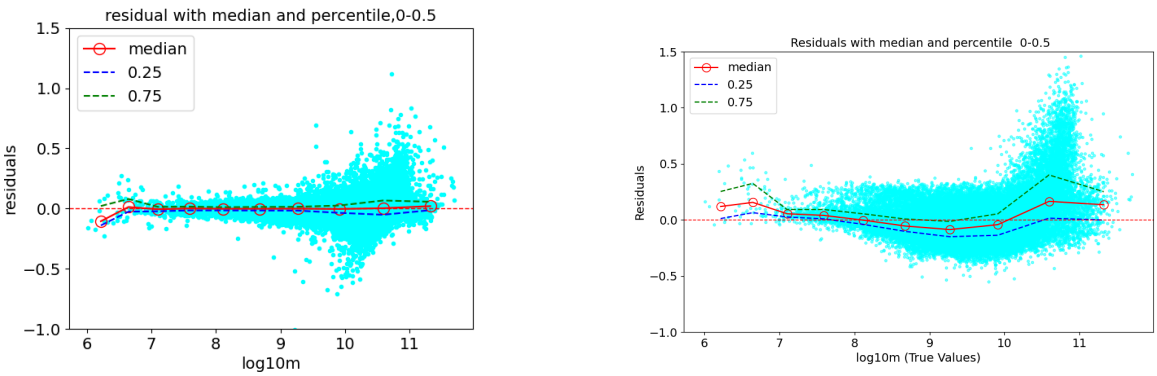
Building upon the robust foundation established by our diagnostics—which identified M_z and color_iz as the dominant features and the redshift range $z < 0.5$ as the optimal performance regime—we proceeded to derive a concise, physically interpretable formula for stellar mass.

3.5.1 Final Feature Selection and Symbolic Regression

Guided by the Random Forest feature importance ranking 7a, we selected the top five photometric parameters for symbolic regression: M_z, color_iz, color_ug, color_gr, and color_ri. This set captures the essential luminosity and stellar population information while excluding redundant or low-impact features. A Random Forest model trained solely on this refined feature set within the $z < 0.5 range$ already achieved an exceptional performance (Test $\mathbf{R^2} = 0.992$, RMSE $= 0.067$ dex, 10a), confirming its sufficiency.

he symbolic regression process explored formulas of increasing complexity. The results demonstrated a clear trade-off: beyond a certain point, added complexity yielded diminishing returns in loss reduction. The most parsimonious model that captured the essential physics without overfitting was found at complexity 9, with the formula: $y = (x0 * -0.3352) + 2.4551 + (x1 * 3.5274)$, where y = $\log_{10} M$, x0 = M_z, and x1 = color_iz.

This model achieved a test R^2 of 0.958 and an RMSE of 0.152 dex. While this represents a slight performance decrease compared to the full Random Forest model, it is a deliberate and favorable trade-off for a vast gain in interpretability and simplicity, resulting in a practical, equation-based estimator.A direct comparison of the residual patterns between the full Random Forest model and the final symbolic regression formula (10) provides a clear visual summary of the trade-off between predictive power and interpretability.



(a) Random Forest (Test $R^2 = 0.992$, RMSE = 0.067).

(b) Symbolic regression (Test $R^2 = 0.958$, RMSE = 0.152).

Figure 10: Direct comparison of residuals (predicted−true). Percentile lines (25/50/75th) overlaid.

As expected, the Random Forest model exhibits superior predictive accuracy. Its residuals are more tightly clustered around the zero line, and the 25th, 50th, and 75th percentile lines run almost perfectly parallel to and centered on zero across the entire mass range. This performance is attributable to two main factors:

1. Feature Utilization: The Random Forest model leverages the complete set of 13 photometric and redshift features, allowing it to capture subtle, non-linear relationships that a simple linear combination of two variables cannot.

2. Algorithmic Advantage: As an ensemble method, Random Forest naturally models complex interactions between variables and is highly robust to overfitting, which contributes to its minimal bias and low variance.

In contrast, the symbolic regression formula shows a slightly wider scatter of residuals and a minor but consistent offset of the median percentile line at the very high-mass end. This is the anticipated cost of simplicity. By distilling the relationship down to a mere two primary features (M_z and color_iz), the formula necessarily omits some of the finer-grained information used by the Random Forest. However, it is critical to emphasize that the symbolic regression residuals still demonstrate excellent behavior: they are unbiased for the vast majority of the mass range and maintain a stable variance. The slight performance gap is a deliberate and justified compromise for achieving a compact, physically transparent, and easily deployable analytical tool.

3.5.2 The Final Formula and Its Physical Interpretation

The final, recommended formula for estimating the stellar mass of galaxies in the low-redshift universe ($z < 0.5$) is:

$$\log_{10} M = M_z * (-0.33520013) + color_{iz} * 3.5273592 + 2.4550812 \quad (2)$$

1. -0.335 × M_z: The z-band absolute magnitude serves as a direct proxy for the total stellar light, with the negative coefficient correctly indicating that more luminous (less negative M_z) galaxies have higher masses. The coefficient's value reflects the mass-to-light ratio in the z-band.

2. $+3.527 \times color_iz$: The i-z color acts as a critical corrective term. A redder color (larger i-z) generally indicates an older stellar population and/or higher metallicity, which corresponds to a higher mass-to-light ratio. This term fine-tunes the mass estimate based on the nature of the stellar population, moving it beyond a simple luminosity measurement.

3. $+ 2.455$: This is the normalization constant that calibrates the equation to the stellar mass scale.

3.5.3 Validation of the Final Model

The reliability of this final formula is rigorously validated through residual analysis (11,12).

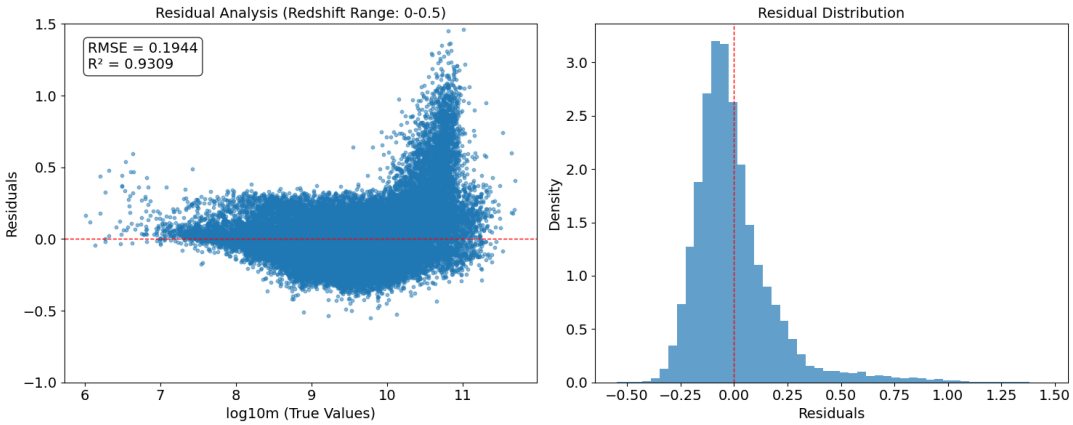


Figure 11: Symbolic Regression Residual Diagnosis. Left: Scatter of residuals versus predicted values shows random, unbiased distribution around zero. Right: A histogram of the residuals confirms their approximately normal distribution.

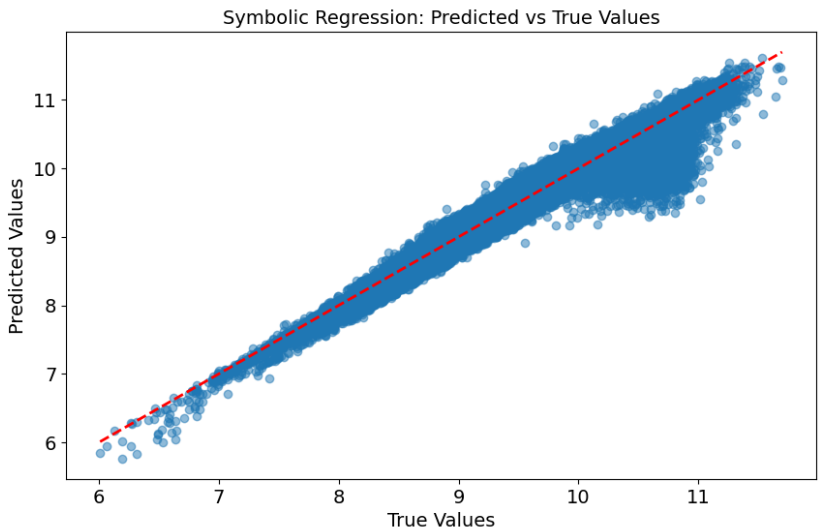


Figure 12: Symbolic Regression Prediction Fit. The final formula's predictions (red line) are plotted against the true stellar mass values (blue points), demonstrating an excellent fit to the data.

Figure11 provides a diagnostic view of the residuals in the low-redshift regime ($0 < z < 0.5$). The left panel shows residuals as a function of the true stellar mass. The scatter remains tightly clustered around zero for intermediate masses, but broadens at the high-mass end where photometric features provide weaker leverage. The right panel displays the residual distribution, which peaks sharply at zero but exhibits a noticeable positive tail. This indicates that while the estimator is unbiased on average, it tends to slightly over-predict the masses of the most massive galaxies. Such systematic departures at the high-mass end are expected given the rarity of bright galaxies and the impact of photometric errors.

Figure 12 compares the symbolic regression predictions directly to the true stellar masses. The points are tightly aligned along the one-to-one line, demonstrating that the compact analytical formula captures the bulk of the variance in the data. Deviations become visible at the lowest and highest masses, consistent with the residual analysis in Fig. 11. Nonetheless, the global performance ($R^2 \approx 0.96$, RMSE ≈ 0.15 dex) confirms that the symbolic expression provides a remarkably faithful approximation to the more flexible Random Forest model.

In conclusion, we have successfully distilled the complex mapping from photometry to stellar mass into a simple, yet powerful, analytical formula. Derived through a data-driven and physically motivated workflow, this tool provides a highly accurate and interpretable method for estimating galaxy stellar masses, particularly valuable for rapid analysis in large photometric surveys.

4 Conclusions

In this study, we have distilled a complex astrophysical relation into a practical, equation-based estimator for galaxy stellar masses. By systematically diagnosing the feature space with Random Forests and validating the stability of performance across redshift, we established that the local universe ($z < 0.5$) offers the optimal regime for deriving robust mass estimators. The symbolic regression formula based solely on z-band luminosity and $i - z$ color encapsulates the key physics of the stellar mass–light connection while remaining highly interpretable. Although machine-learning ensembles achieve higher predictive accuracy,the simplicity of the analytical formula represents a favorable trade-off, enabling rapid application in large photometric surveys.

Our findings highlight two broader implications. First, the redshift and bandpass dependence of the mass–magnitude relation reflects the interplay between galaxy assembly history, star formation, and stellar population aging, offering a physically motivated rationale for feature selection. Second, the methodology developed here—combining diagnostic machine learning with symbolic regression—demonstrates a generalizable framework for extracting interpretable physical laws from noisy observational data.

Future work could extend this framework in several directions. One natural avenue is to push the symbolic regression analysis to higher redshift domains, where the relation becomes noisier, by combining alternative algorithms or hybrid approaches to mitigate observational uncertainties. Another promising direction is to investigate in greater depth the evolution and interdependence of the dominant features (e.g., M_z, $i - z$, and other colors) across cosmic time, which may reveal additional physical insight into galaxy assembly and stellar population processes. More generally, incorporating spectroscopic constraints or cross-survey comparisons could test the robustness of the derived formula in heterogeneous observational settings. Such extensions would further consolidate the utility of symbolic regression as a bridge between machine learning performance and astrophysical interpretability.

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